

Acoustic Imaging for 2D Tracking of Surface Vessels Using Distributed Acoustic Sensing on Submarine Optical Fibres

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Abstract *This paper presents a beamforming-based method using fibre-optic distributed acoustic sensing to map acoustic energy in two dimensions around submarine telecom fibres. The method enables near-field 2D localisation via beamforming and tracking through Kalman filtering of moving marine vessels, including regions without direct fibre deployment. ©2026 The Author(s)*

Introduction

Distributed acoustic sensing (DAS) [1] measures Rayleigh phase variations in optical fibres to reconstruct a distributed profile of strain induced by acoustic perturbations along thousands of sensing points. Its ability to use existing telecom fibres enables low-cost, wide-aperture sensing for applications such as seismology, oceanography, and maritime surveillance [2,3]. However, DAS is limited to detecting acoustic waves reaching the fibre itself, restricting localisation only along its longitudinal axis and preventing direct two-dimensional (2D) localisation without dense fibre deployments. In addition, trade-offs between sensitivity, bandwidth, and sensing range further limit the achievable 2D spatial coverage.

To overcome these limitations, beamforming approaches can be used to treat the fibre as a distributed array [4-6], exploiting the spatiotemporal features of DAS data for source localisation and signal enhancement. But, many approaches rely on restrictive assumptions (e.g., specific fibre geometries, narrowband signals, far-field conditions), limiting applicability to direction-of-arrival estimation, while neglecting intrinsic DAS characteristics such as non-uniform sensitivity, axial response, and multipath propagation. Recent near-field methods enable 2D localisation through triangulation [6], but are limited to single-source scenarios and degrade in complex environments.

To resolve this, a grid-based beamforming approach was recently proposed [7] for near-field acoustic imaging, mapping acoustic energy over a 2D region. However, this method has only been proven for static sources using long energy calculation windows, limiting dynamic scenarios and preventing the tracking of moving sources.

In this paper, we extend this concept to dynamic scenarios, introducing a DAS-based near-field acoustic imaging method for the localisation and tracking of moving surface vessels using submarine telecom fibres. Source localisation is achieved via beamforming-based acoustic

imaging, generating 2D acoustic energy maps over short time windows (e.g. 2 s), while tracking is performed using a classical Kalman filter [8,9] to ensure smooth motion and mitigate symmetry-induced ambiguities typical of linear sensor arrays, like submarine fibre cables. This approach transforms the fibre into an acoustic imager capable of capturing acoustic source dynamics over large areas, including regions far from the fibre, highlighting its potential for maritime surveillance.

Acoustic imaging and Kalman tracking

Near-field acoustic imaging is performed using delay-and-sum beamforming [10,11] on DAS measurements to reconstruct the acoustic energy distribution around the fibre. The region of interest is discretised into a 2D grid, where each pixel is treated as a potential source [7]. For each pixel, propagation delays to DAS channels are computed from the known fibre geometry and the acoustic velocity (known or estimated from data [7]). The signals from DAS channels are time-aligned and coherently summed, implementing a spatial filter that scans the imaged region and enhances wavefronts from the target location while suppressing others. The acoustic energy is estimated by integrating the squared beamformer output over a time window, while repeating this process across the grid yields an acoustic image.

This approach transforms the DAS-interrogated optical fibre into an acoustic imager [7], enabling monitoring beyond the fibre path without full spatial coverage. The integration window defines the dynamic capability of the approach. Short windows capture fast motion but reduce the accumulated energy and the image contrast. Instead, longer windows improve contrast through higher energy integration but suit only slow or static scenarios. Thus, for dynamic scenarios, the window must be short enough to follow the source motion. To ensure reliable imaging, only DAS channels with high acoustic energy must be

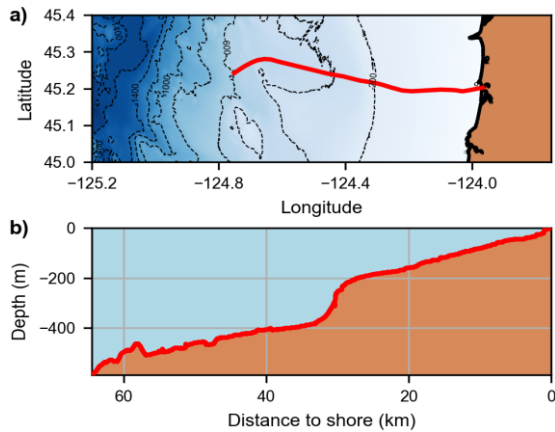


Fig. 1: a) Fibre layout and surrounding area bathymetry. b) Fibre depth, with shore on the right side of the plot.

used, excluding low-energy or noisy channels [7].

The pixel with the maximum acoustic energy is taken as the source location at each frame. To ensure physically consistent motion, the estimated position is tracked over time using a classical Kalman filter [8,9] based on a constant-velocity model, with the state vector including position and velocity. The filter recursively predicts and updates the source position from the image-based estimations, smoothing the source trajectory and reducing errors due to noise and ghost peaks, thus enabling robust and continuous tracking.

Submarine DAS measurement conditions

The DAS dataset used in this study, reported by Wilcock & OOI [12], is acquired using two submarine fibres off the coast of Oregon (northern Pacific Ocean). Here, to validate the proposed method, only DAS data from the northern cable is used, providing a sensing distance of ~ 64.8 km, as shown in Fig. 1a, limited by the presence of an optical repeater at that position. The fibre is buried ~ 1.5 m below the seafloor and reaches water depths of up to ~ 500 m, as shown in Fig. 1b.

DAS measurements are collected during a scheduled shutdown using commercial interrogators. The proposed method is validated on measurements from a 180 m-long vessel detected at a ~ 48.7 km distance along the cable (~ 45.5 km from shore). The DAS system is set with a 500 Hz sampling rate, 2 m channel spacing and a 30 m gauge length. To reduce computational load and avoid the use of highly correlated spatial points, the data is downsampled to 10 m spacing, yielding $\sim 6,480$ sensing points along the fibre.

Experimental Results

Fig. 2 shows the waterfall plot (distance versus time) of the acoustic signal along the northern optical fibre during the vessel transit. Significant acoustic energy is observed within ± 1 km of the crossing point, while channels farther away show weak or no response. A reduced response is also visible directly below the surface vessel, possibly

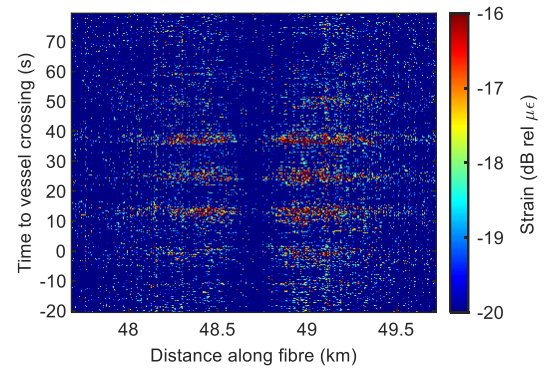


Fig. 2: Waterfall plot of DAS measurements during the passage of a 180 m-long surface vessel crossing the fibre.

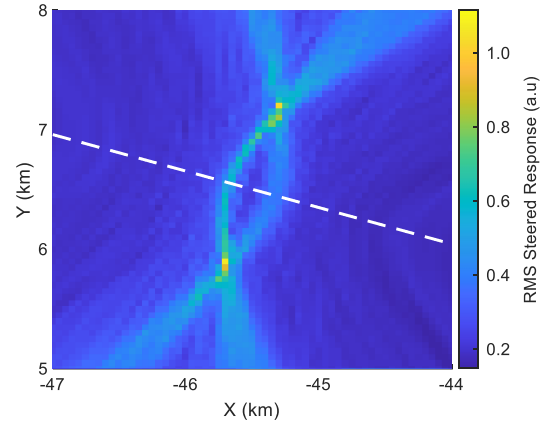


Fig. 3: Acoustic image obtained 127 s after the crossing, using DAS data within ± 1 km of the crossing point

due to the absence of axial strain at near-90° incidence, whereas lateral fibre sections exhibit stronger coupling with the acoustic wave. The absence of longitudinal propagation indicates that the vessel crosses nearly perpendicular to the fibre. As the waterfall representation does not provide information about the vessel's distance to the fibre, delay-and-sum beamforming is used to generate acoustic images over a 3×3 km area centred at the crossing point, with a 50×50 m resolution. Using 200 channels over 2 km and 2 s time windows, the acoustic energy is mapped across the grid, as shown in Fig. 3. The source is identified by the upper brightest yellow pixel. However, a secondary ("ghost") peak (lower yellow peak) also appears on the opposite side of the fibre (dashed white line), due to the linear DAS array geometry typical of most submarine fibre cables, introducing localisation ambiguity. This ghost peak disappears in non-straight fibres, where the fibre deployment geometry introduces angular diversity that resolves the ambiguity [7].

Using overlapping windows of 2 s, acoustic images are generated at a 1 s refresh rate. Fig. 4 shows selected frames before and after the crossing, comparing the peak-based localisation (yellow-filled red circles) with the actual vessel positions obtained from Automatic Identification System (AIS) data (cyan dots) [13]. Note that, in some cases (e.g. Fig. 4a and 4d), noise causes

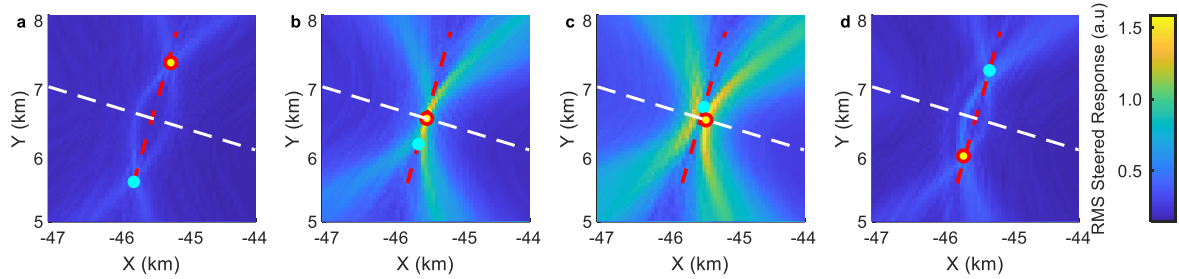


Fig. 4: Beamforming-based acoustic images from DAS measurements. Estimated vessel positions (red circles) are compared with AIS data (cyan dots) at: a) -149 s, b) -61 s, c) +27 s and d) +115 s relative to the crossing.

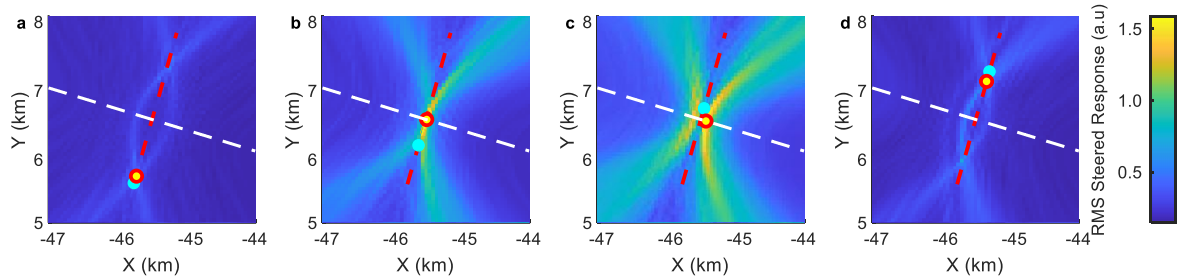


Fig. 5: Beamforming-based acoustic images from DAS measurements. Estimated vessel positions after Kalman filtering (red circles) are compared with AIS data (cyan dots) at: a) -149 s, b) -61 s, c) +27 s and d) +115 s relative to the crossing.

the ghost peak to dominate, leading to large localisation errors, as a simple peak detection alternates between the two symmetric solutions.

To resolve this localisation ambiguity, a Kalman filter is applied to track the source position over time, assuming a constant vessel speed. Fig. 5 shows the same acoustic images as Fig. 4, but with source localisation obtained after Kalman filtering, enabling consistent identification and tracking of the true source. Note that the filter requires only minimal prior information, such as the initial position or the direction of travel, to secure correct tracking. In this case, this information can be obtained from the DAS measurements on the second optical fibre. Results demonstrate that the Kalman filter effectively suppresses spurious transitions between symmetric (real and ghost) peaks, significantly improving the localisation accuracy at a temporal resolution of 2 s.

Fig. 6 shows the localisation error between the estimated vessel position and the AIS data. Results confirm that Kalman filtering removes large errors caused by the peak ambiguity, with a minimum error of 110 m at the crossing point, where the acoustic energy is highest, and true and ghost peaks coincide. The error increases with distance but shows a secondary reduction at larger vessel positions from the fibre. This behaviour may be explained by the beamformer response: at large distances, the array response corresponds to two intersecting parabolic curves that yield a well-defined maximum, while closer to the fibre their intersection broadens the high-energy region, reducing localisation precision. Near the fibre crossing, however, the high-energy region becomes again more localised while the acoustic energy reaching the fibre increases, leading to an improved localisation accuracy.

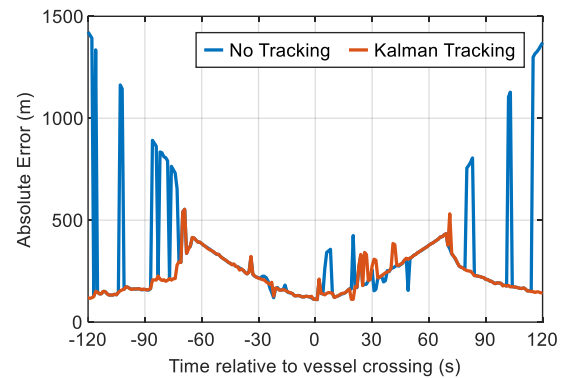


Fig. 6: Absolute localisation error of the vessel position, without and with Kalman-based tracking.

Conclusions

In this paper, we have proposed and validated a method for localising and tracking moving acoustic sources by combining beamforming-based acoustic imaging with Kalman filtering applied to DAS measurements on existing submarine telecom fibres. A 180 m-long surface vessel has been successfully tracked, achieving localisation errors as low as 110 m. Note that, in straight fibre deployments (typical of most submarine cables), the acoustic images exhibit a ghost source due to symmetry ambiguity inherent to linear arrays. The Kalman filter resolves this ambiguity by enforcing smooth motion and tracking the true source trajectory. Further work is needed to better understand the factors limiting accuracy, including the DAS array response, emitted acoustic energy, and uncertainties in the fibre positioning. In addition, an optimisation of the temporal integration window, based on the vessel speed, could increase the accumulated acoustic energy, enhance image contrast and reduce localisation errors.

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